

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and deployed **Java Spring Boot** microservices for an enterprise client, handling complex business logic and integrating with existing systems.
- Implemented data persistence layers using **Hibernate** and **JPA**, managing complex object-relational mappings and optimising database interactions for critical financial applications.
- Designed and implemented object-oriented solutions in **Java**, adhering to design patterns and ensuring code quality through comprehensive **JUnit** testing and **Mockito**.
- Contributed to the design and implementation of **smart energy services**, focusing on energy efficiency measures and integration with renewable energy sources.
- Led research tasks within EU-funded projects, focusing on **energy performance contracting** and developing methodologies for energy efficiency in commercial and industrial sectors.
- Collaborated with external partners on **energy policy analysis** and regulatory innovation, contributing to reports that informed new policy development for sustainable energy.
- Containerised applications using **Docker** and deployed them to **AWS EC2** instances, managing configuration and ensuring smooth rollouts via **Jenkins** pipelines.

SKILLS

Programming Languages – Java, Python, SQL, R

Frameworks & Libraries – Spring Boot, Hibernate, JPA, pandas, NumPy, scikit-learn, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, RStudio

Methodologies – Agile, Scrum, CI/CD

Domain Expertise – Smart Energy Services, Energy Efficiency, Energy Policy, Regulatory Innovation, Energy Performance Contracting

PROJECTS

Smart Energy Service Integration Platform (HCL Technologies)

- Developed a **Java Spring Boot** application to integrate data from various smart meters and energy sensors, enabling real-time monitoring of energy consumption patterns.
- Implemented RESTful APIs for data ingestion and retrieval, facilitating seamless communication between the platform and downstream analytics services for energy efficiency analysis.
- Designed and implemented database schemas in **PostgreSQL** to store time-series energy data, supporting efficient querying for historical performance analysis and reporting.

Energy Efficiency Measure Recommendation Engine (MSc Applied Project)

- Built a **Python**-based recommendation engine using **scikit-learn** to suggest energy efficiency measures for residential buildings based on historical consumption data and building characteristics.
- Performed exploratory data analysis and feature engineering using **pandas** and **NumPy** on a dataset of 15,000 building profiles, identifying key factors influencing energy usage.
- Developed interactive **Power BI** dashboards to visualise the impact of recommended measures and present findings to project stakeholders, demonstrating potential energy savings.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)